Comprehensive Analysis and Prediction of Air Quality Index Using Statistical and Machine Learning Techniques

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***Abstract*—Air Pollution is an environmental and health problem which affects millions of people globally. This paper suggests a methodological approach to the analysis and prediction of Air Quality Index using the combination of statistical techniques and machine learning algorithms. The methodology comprises several stages of data analysis, namely: data cleaning, outliers removal, normalization, EDA, and predictive modeling.**

Various visualization techniques such as histograms, scatter plots, bar chart, heatmaps are used to understand pollution distribution, relationship between different variables. A linear regression algorithm was used to predict air pollution depending on the set of selected features.

The findings show that the model demonstrates high performance as indicated by the R^2 equaling to around 0.92. This study highlights the importance of performing necessary preprocessing procedures and using simple machine learning algorithms in order to obtain reliable results.

Recently, environmental monitoring has gained much importance due to the rapid development of environmental science, increasing amounts of information related to it. Machine learning algorithms are now widely used for environmental data analysis in order to detect trends, build models and make predictions.

As for traditional statistical methods, they can hardly handle huge volumes of data and perform necessary calculations effectively. Simple machine learning algorithms combined with proper data analysis allow performing the necessary tasks successfully.

***Index Terms*—Air Quality Index, Machine Learning, Linear Regression, Environmental Monitoring, Data Analysis**

I. Introduction

Air pollution represents a key environmental problem facing us today. The development of industry, urbanization, and the increase in exhaust gas emissions have led to significant deterioration in the quality of the air in many areas. Poor air quality impacts negatively on people's health by causing heart and breathing-related illnesses and even early death.

AQI stands for Air Quality Index. It is an effective measurement tool used to estimate the level of air pollution as well as possible adverse effects on human health. Effective air pollution analysis and AQI forecast are important issues from both research and practical perspectives.

The main goal of the current study is to create a model aimed at efficient pollution analysis and AQI forecasting using machine learning algorithms.

Contributions:

- Creating a comprehensive pipeline with preprocessing steps that handle missing values and outliers
- Performing detailed exploratory data analysis to understand the patterns and distribution of pollution
- Using normalization techniques to normalize the variables/features
- Building a linear regression model to predict AQI
- Analyzing results and understanding their practical significance

II. Dataset Description

The data set used in this research involves air pollution data that were recorded in various states and cities of India through the Government of India open data portal (data.gov.in). The data set includes three main attributes, namely pollutant_min, pollutant_max, and pollutant_avg.

This indicates the range and mean values of the pollutants in various geographical locations. This data set involves spatial data on the extent of pollution levels across various locations.

This means that AQI scores are highly associated with human health conditions. The higher the pollutant level, the higher the likelihood of developing respiratory and heart diseases. As a result, analyzing this feature becomes necessary for making decisions.

However, it should be noted that this data set does not contain any temporal data and meteorological attributes such as temperature, humidity, and wind speed.

III. Data Preprocessing

A. Missing Value Handling

Missing values were handled using mean imputation:

$$x_{filled} = \frac{1}{n} \sum x_i \tag{1}$$

This approach preserves the overall distribution of data without introducing extreme bias.

B. Outlier Removal

Outliers were removed using the Interquartile Range (IQR) method:

$$IQR = Q3 - Q1 \tag{2}$$

$$Lower = Q1 - 1.5IQR, \quad Upper = Q3 + 1.5IQR \tag{3}$$

Data Preprocessing Step: The data preprocessing step is necessary for guaranteeing good-quality data. Outliers could lead to distorted statistics and compromise the functioning of models. Removing outliers facilitates consistent analysis.

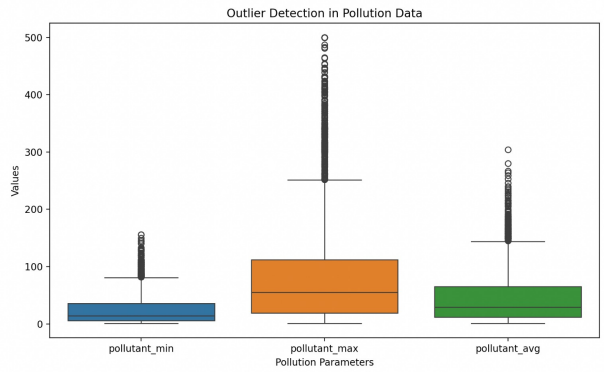

Fig. 1. Boxplot Before Outlier Removal

Description: This chart shows that there are extreme values in the pollution data set, which could lead to problems with the analysis because of possible anomalies.

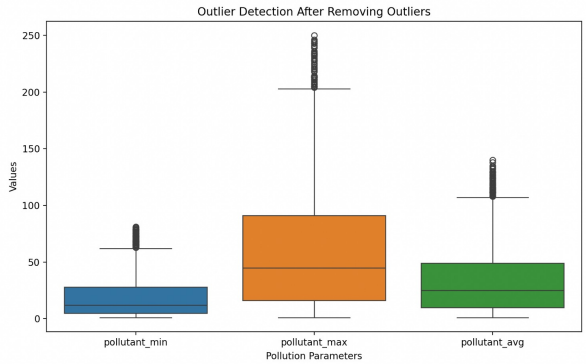

Fig. 2. Boxplot After Outlier Removal

Description: The removal of the extreme values through the use of the IQR approach leads to the generation of a more stable data set for further analysis.

IV. Exploratory Data Analysis

A. State-Level Analysis

It is clear from the analysis that there is uneven pollution across different regions. Some states tend to be more polluted, which is determined by their environment and industries.

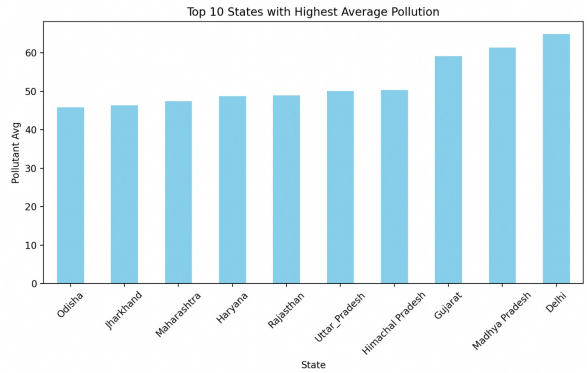

Fig. 3. Top Polluted States

Description: The graph shows that pollution occurs in only a few states, suggesting environmental imbalances in different regions.

B. City-Level Analysis

There is a lot more pollution in urban areas compared to the countryside. The reason for this is due to heavy traffic, industries, and increased population.

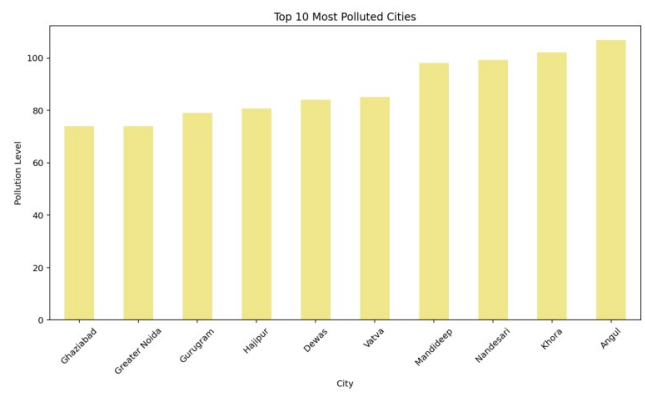

Fig. 4. Top Polluted Cities

Description: There is very high pollution in urban settings due to high concentration of people, vehicles, and industries.

C. Distribution Analysis

It can be noticed that the graph is skewed to the right, indicating that the instances of very high pollution are localized in certain regions. Regions that have experienced high minimum levels of pollution also report high peak levels of pollution.

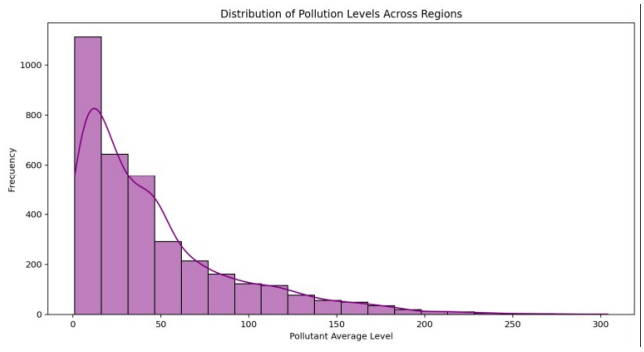

Fig. 5. Distribution of Pollution Levels

Description: The skewness to the right implies that highly polluted areas exist in certain locations.

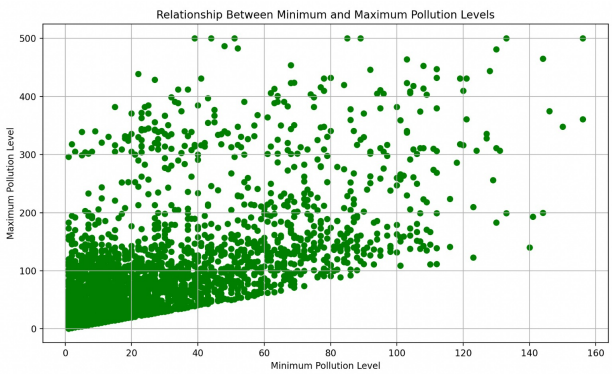

Fig. 6. Relationship Between Minimum and Maximum Pollution Levels

*Description:*This scatter diagram shows that there is a positive association between minimum and maximum pollution values.

D. Correlation Analysis

A strong correlation between pollutant features indicates consistent pollution patterns. To further understand relationships between variables, correlation analysis is performed.

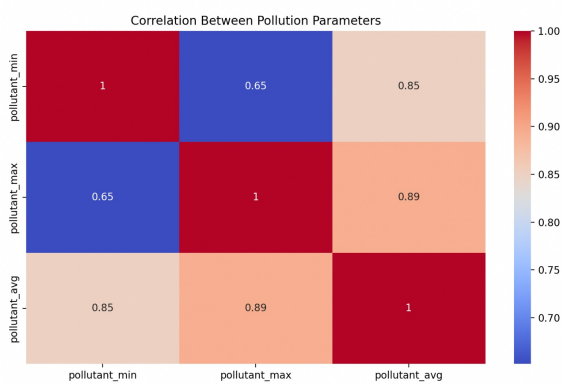

Fig. 7. Correlation Between Pollution Variables

Description: A strong positive correlation between pollutant variables suggests consistent pollution patterns across various regions.

V. Statistical Insights

- The pollution indicators are geographically concentrated instead of having an equal distribution in all the cities and states.
- There are positive correlations between the pollutants, such as pollutants_min, pollutants_max, and pollutants_avg.
- The presence of outliers in the pollution dataset significantly impacts its overall distribution.

VI. Feature Scaling

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}} \tag{4}$$

Feature scaling allows for equal contribution of all variables while minimizing bias in the training process. In the experiment, Min-Max Scaling was used for scaling features to the range of [0, 1] before training the Linear Regression model.

VII. Experimental Setup

The data was divided into the Training Set (80%), and Test set (20%). Implementation of Linear Regression Model is done using Scikit-learn in Python.

Performance metrics used:

- Mean Squared Error (MSE)
- Mean Absolute Error (MAE)
- R-Square Score (R^2)

VIII. Implementation Details

The design was coded using the Python programming language, with several libraries used to help with data manipulation, visualization, and modeling processes. Data manipulation and preparation were performed using Pandas, while NumPy helped with the numerically intensive tasks. Visualizations were done using Matplotlib and Seaborn to glean insights from the data. The Linear Regression model was built using Scikit-learn, along with other machine learning models.

The process was done in a systematic fashion following the steps of data cleaning, preparation, scaling, modeling, and evaluation. Careful attention was given at each step to guarantee consistent results.

The experiments were carried out on a typical computer system lacking special hardware like GPUs, thus showcasing the method's efficiency and applicability in practical settings where computational resources may be limited.

IX. Model Development

A Linear Regression model was used to predict pollution levels:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 \tag{5}$$

The model uses *pollutant_min* and *pollutant_max* as input features and predicts *pollutant_avg* as the target variable.

Linear regression is simple and understandable, making it suitable for explaining the interdependence between pollution variables. Linear regression was built using normalized data in order to improve its efficiency.

X. Results and Analysis

Linear regression shows impressive results when predicting, as the bulk of data falls near the reference line, suggesting a high degree of accuracy.

The model showed an R^2 value of about 0.92, meaning that the variance of the pollution data is captured significantly. MAE is relatively small, suggesting low errors in predictions.

However, there are slight errors for more pollution data, which is expected since it reflects reality.

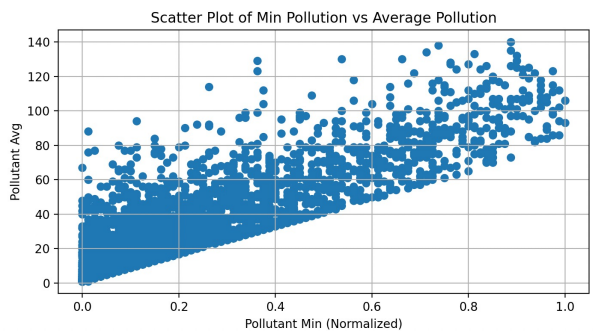

Fig. 8. Relationship Between Minimum and Average Pollution

Description: The scatter plot shows a strong positive relationship between minimum and average pollution levels.

As minimum pollution increases, the average pollution also rises, indicating consistent pollution patterns across regions rather than random fluctuations.

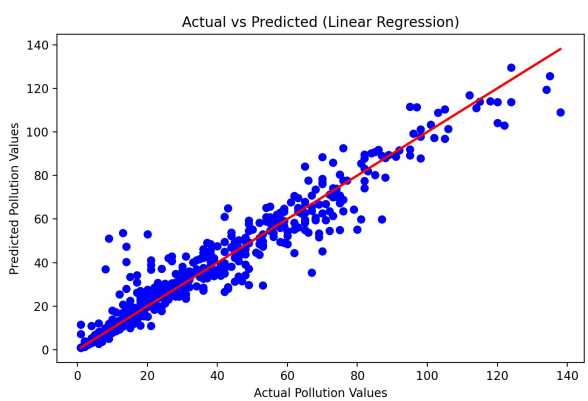

Fig. 9. Actual vs Predicted Values

Description: The scatter plot compares actual and predicted values, showing how closely the model fits the data.

XI. Detailed Interpretation of Results

The accuracy of machine learning models in forecasting AQI demonstrates the complexity of environmental information. The algorithm Linear Regression assumes a linear correlation between predictors and dependent variables, which is relevant in capturing trends in pollution data.

Environment is influenced by several interrelated factors, including pollution from industrial activities, number of vehicles, and meteorological phenomena. As a consequence, there appears some deviation in pollution values that does not follow the linear trend.

Yet, the model shows great prediction accuracy due to a high coefficient of determination (R^2) and relatively small values of error measures. The study reveals that a fairly simple model like Linear Regression can provide satisfactory forecasts if applied along with preprocessing algorithms.

Thus, the study proves that choosing the right machine learning model and performing adequate data preprocessing is a crucial step in predicting environmental values.

XII. Model Comparison

TABLE I
Model Performance Comparison

Model	MSE	MAE	R^2
Linear Regression	57.97	4.54	0.9271

The output shows that the Linear Regression model possesses high predictive ability in predicting Air Quality Index (AQI) values. The high value of R^2 shows that the model captures a large percentage of the variance in the data, and the low value of MAE signifies the low error in the predictions made by the model.

Despite being based on assumptions of linearity in the variables used, the success of Linear Regression depends on proper data preparation and scaling. From the results, we observe that even the simplest model can yield good results when applied to preprocessed data.

From the above discussion, the results show that Linear Regression is effective in predicting AQI in the current study.

XIII. Key Insights

- Pollution is driven by regional and urban factors
- Strong relationships exist between pollutant variables such as pollutant_min, pollutant_max, and pollutant_avg
- Data preprocessing and feature scaling significantly improve model performance

XIV. Discussion

The outcomes of the current study highlight that air pollution patterns demonstrate structural characteristics rather than randomness and are influenced by structural and environmental

drivers. Urbanization, industrial processes, and population density are identified as important pollution drivers.

Linear regression represents a relatively simple yet intuitive prediction method capable of capturing general tendencies observed in environmental data. However, this algorithm may prove inefficient in cases of non-linear inter-variable relations, especially at pollution extremes.

The outcomes emphasize the significance of adequate data preprocessing techniques, such as outliers elimination and normalization, in improving the efficiency of the prediction process. Data preprocessing techniques significantly increase the reliability of the prediction model.

Thus, the results reveal that a relatively simple ML model can produce relevant outcomes in case of adequately prepared data.

XV. Environmental Impact Perspective

Air pollution affects the vulnerable population adversely and poses considerable public health threats. The application of data-based predictive models such as AQI analysis through machine learning techniques can prove helpful in determining risk-prone areas and making sound environmental policies.

XVI. Limitations

- Lack of temporal data for analyzing trends over time
- Absence of meteorological variables such as temperature, humidity, and wind speed
- Limited feature set used for model training

XVII. Future Work

- Implementation of advanced machine learning models such as XGBoost
- Integration of meteorological data including temperature, humidity, and wind speed
- Application of time-series forecasting techniques for trend analysis

XVIII. Conclusion

This study presents a comprehensive framework for analyzing and predicting Air Quality Index (AQI) using data-driven techniques. The results demonstrate that Linear Regression can effectively model pollution levels when combined with proper data preprocessing, including outlier removal and feature scaling.

The model achieved strong performance with a high R^2 score and low error values, indicating reliable prediction capability. The study highlights the importance of data cleaning, visualization, and simple machine learning techniques in environmental data analysis.

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